# PMP100N08

100 V N-Channel Power MOSFET - SYNTHETIC TEST DATASHEET

## 1. Absolute Maximum Ratings

TA = 25 C unless otherwise noted. (Tj = 125 C maximum continuous)

| Parameter | Symbol | Value | Unit |
|---|---|---|---|
| Drain-Source Voltage | VDS | 100 | V |
| Continuous Drain Current | ID | 25 | A |
| Maximum Junction Temperature | TJ | 175 | C |

## 2. Electrical Characteristics

TA = 25 C unless otherwise noted.

| Parameter | Symbol | Min | Typ | Max | Unit | Conditions |
|---|---|---|---|---|---|---|
| Drain-Source On-Resistance | RDS(on) | - | 5.5 | 7 | mOhm | VGS = 10 V, ID = 10 A, TJ = 25 C |
| Gate-Source Threshold Voltage | VGS(th) | 2 | 3.5 | 4.5 | V | ID = 250 uA |
| Gate-Source Voltage | VGS | - | - | 20 | V | |
| Total Gate Charge | QG | - | 62 | - | nC | VGS = 10 V, ID = 10 A |
| Gate-Drain Charge | QGD | - | 12 | - | nC | VGS = 10 V, ID = 10 A |
| Input Capacitance | CISS | - | 4900 | - | pF | VDS = 50 V, VGS = 0 V, f = 1 MHz |
| Output Capacitance | COSS | - | 900 | - | pF | VDS = 50 V, VGS = 0 V, f = 1 MHz |
| Thermal Resistance, Junction-to-Case | RthJC | - | 1.2 | - | C/W | |
| Reverse Recovery Charge | Qrr | - | 120 | - | nC | ID = 10 A, di/dt = 100 A/us |